Are Birth Rates Really Declining?

A Comparison of Nebraska and National Birth Trends (2020–2024)

Linh Huynh

Bellevue University

DSC680-T301 Applied Data Science (2263-1)

Prof. Amirfarrokh Iranitalab

December 21, 2025

Business Problem

Hospitals and healthcare systems must plan maternity services carefully, including staffing levels, bed capacity, prenatal and postnatal programs, and neonatal intensive care unit (NICU) resources. Decisions are often influenced by national reports suggesting that birth rates in the United States are declining. If these reports are applied without considering local or state-level differences, hospitals may reduce services prematurely or fail to prepare for continued demand.

This project examines whether Nebraska's birth trends follow national patterns and whether recent data supports the idea of a continued decline. The goal is to support data-driven planning decisions for maternity services by using predictive and advanced analytical methods rather than relying on national averages alone.

Background / History

In recent years, media and public discussions have emphasized declining birth rates across the United States. At the same time, anecdotal observations, including reports from healthcare workers and personal experience, suggest that some local hospitals continue to see strong maternity demand. These conflicting signals raise important questions about how national trends translate to state-level realities.

Birth data collected by the Centers for Disease Control and Prevention (CDC) provides an opportunity to examine these patterns systematically. By comparing Nebraska with national trends from 2020 to 2024, this project explores whether Nebraska mirrors national behavior or shows a different pattern that could affect healthcare planning.

Data Explanation (Data Preparation & Description)

Data Source

The dataset used in this project comes from the CDC WONDER Natality Database. It contains aggregated annual birth data by state and maternal characteristics.

Time Period

2020–2024 (5 years)

Data Preparation Steps

The following data cleaning steps were completed:

- Column names were standardized for consistency
- Blank rows were removed
- Data types were converted (strings to numeric values)
- Year and code fields were converted to integers
- Birth counts and percentage fields were converted to numeric format

- Non-analytical metadata fields (such as ‘notes’) were removed

Key Variables Used

- Births (target variable)
- Year
- State of residence
- Census region
- Average age of mother
- Mother’s education level
- Average number of prenatal visits
- Average pre-pregnancy BMI
- Average birth weight

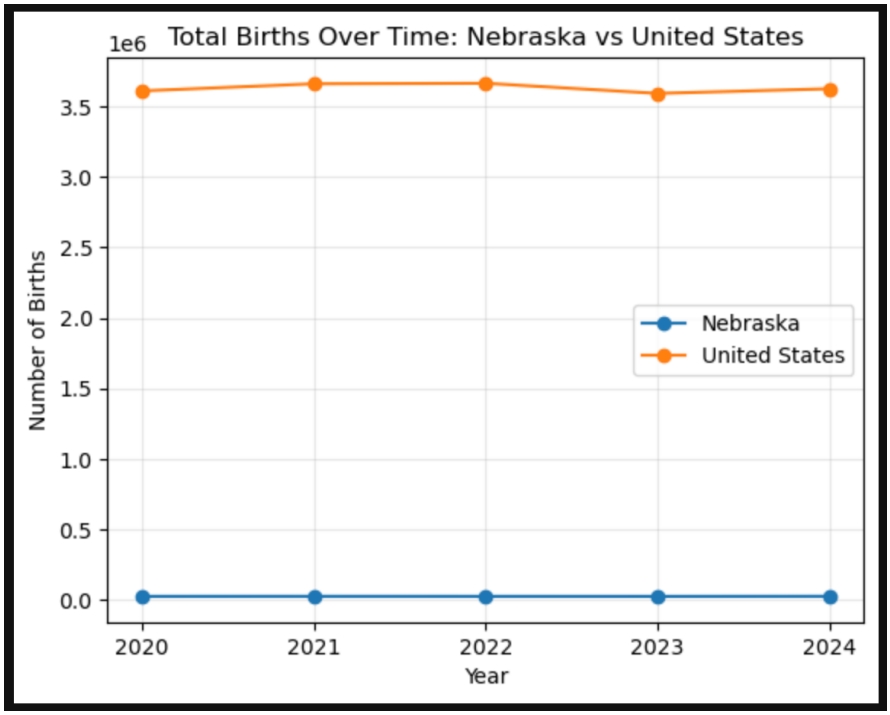
Methods

Methods used in this project include aggregation of birth counts by year, state, and region, and comparative trend analysis between Nebraska and the United States. Demographic composition analysis was conducted by mother’s education level, and correlation analysis was used to examine relationships among key predictors.

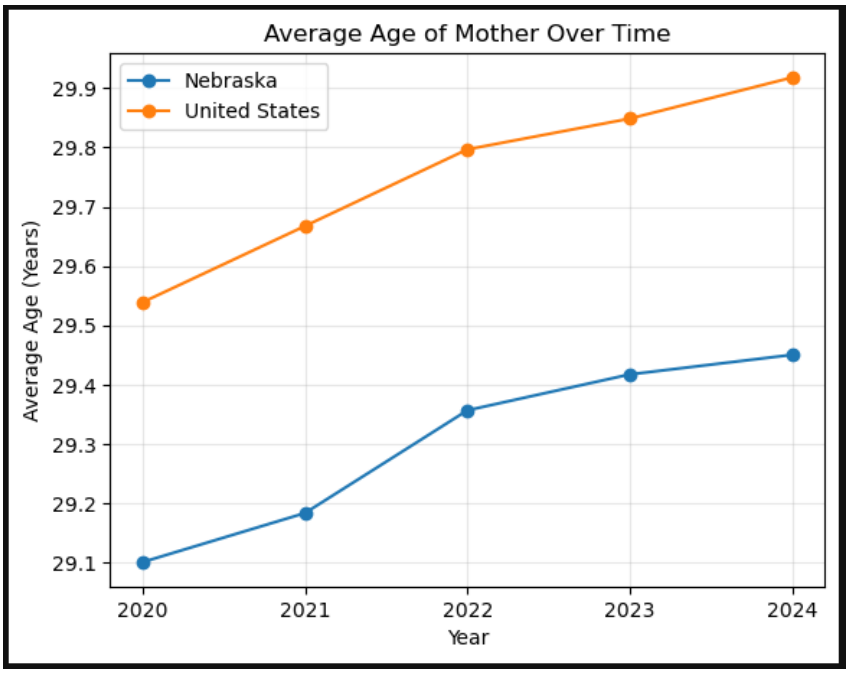
Visualizations were used to guide feature selection and identify patterns that support predictive modeling. Because the dataset includes multiple years, the analysis extends beyond

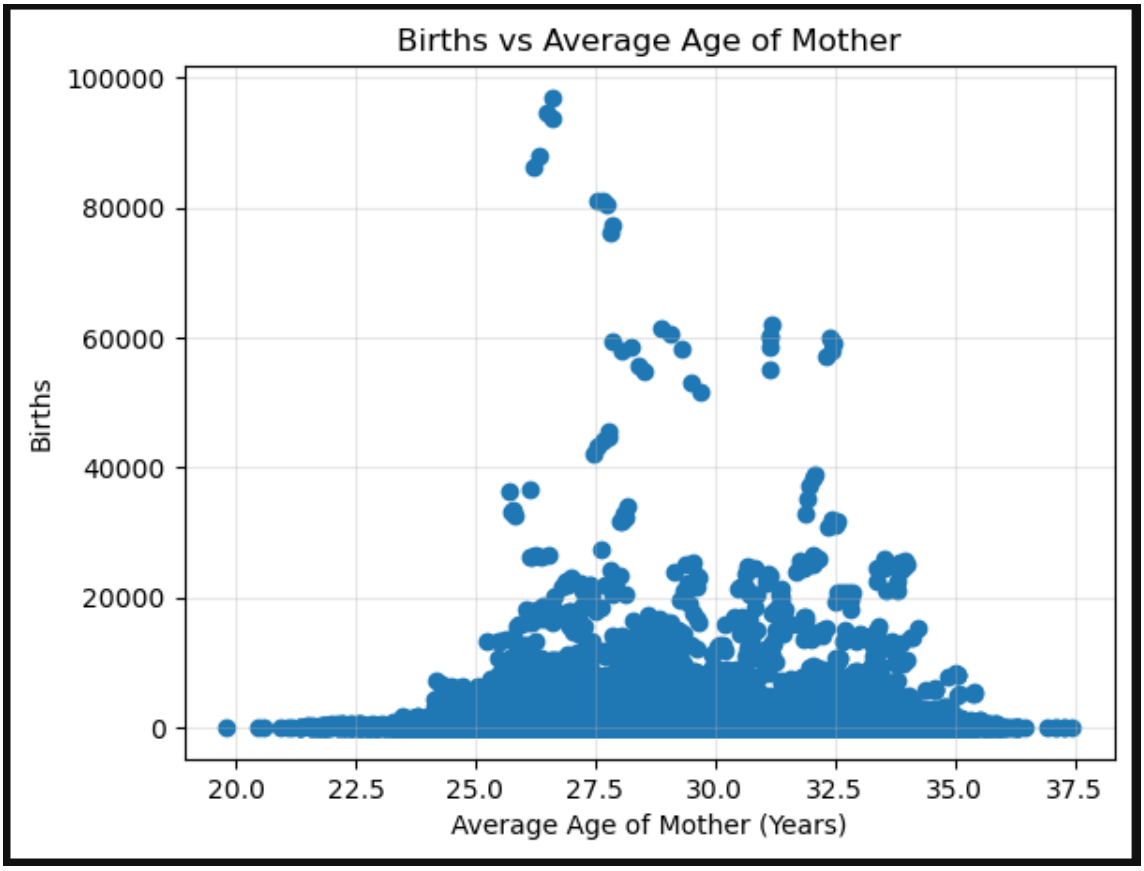
descriptive exploration to include short-term prediction. A baseline linear regression model was developed for interpretability, followed by an advanced random forest model to capture nonlinear relationships and improve prediction accuracy. Model performance was evaluated using standard error metrics to compare results.

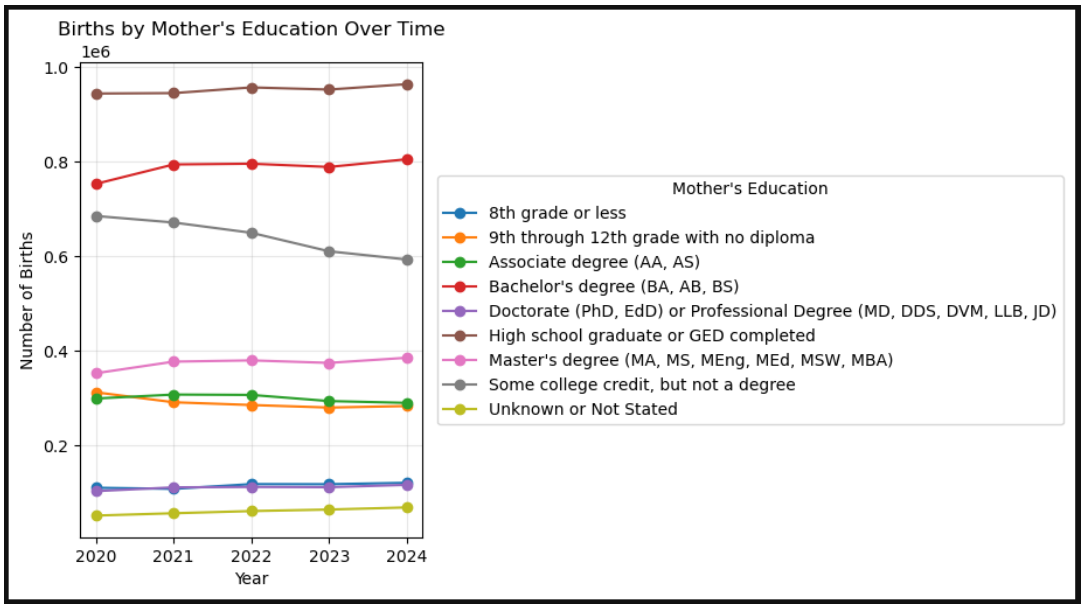
Analysis



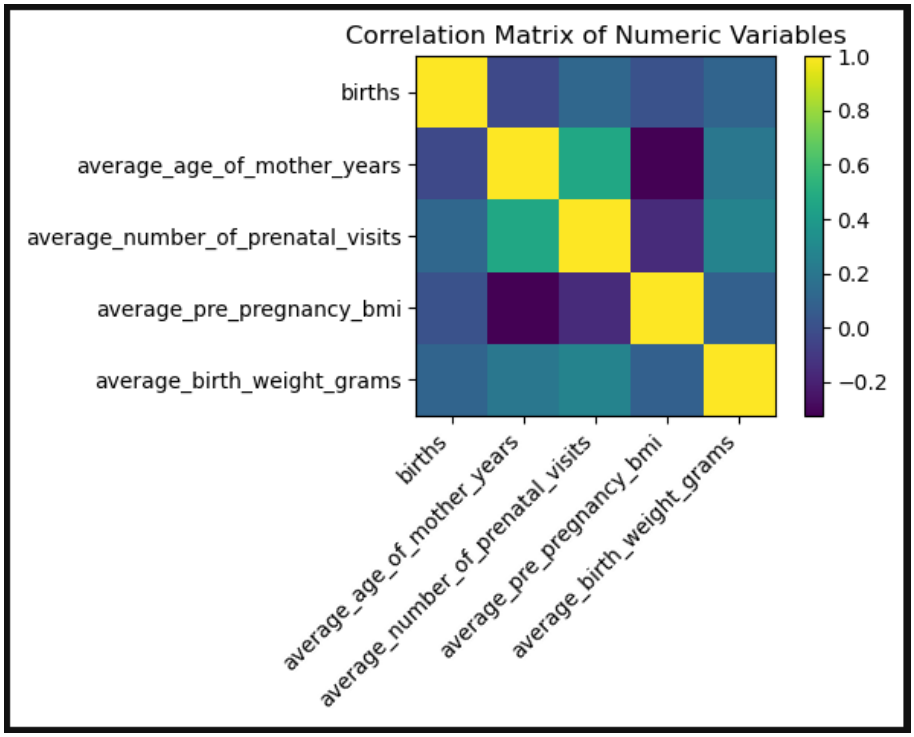
Visual analysis shows that total births in both the United States and Nebraska remain relatively stable from 2020 to 2024. Births increase slightly through 2022, decline in 2023, and recover in 2024. Nebraska follows the same pattern as the national trend, suggesting that state-level birth changes are influenced by broader national factors.


Average Age of Mother Over Time
Nebraska
United States
Average Age (Years)
29.9
29.8
29.7
29.6
29.5
29.4
29.3
29.2
29.1
2020
2021
2022
2023
2024
Year


Births vs Average Age of Mother
Births
100000
80000
60000
40000
20000
0
20.0
22.5
25.0
27.5
30.0
32.5
35.0
37.5
Average Age of Mother (Years)



Maternal characteristics help explain these patterns. The average age of mothers increases steadily over time, and births shift toward higher education levels, with growth among mothers holding bachelor's and master's degrees. These trends reflect delayed childbearing and changing demographics rather than a sharp decline in fertility.



	Model	MAE	RMSE
0	Linear Regression	2036.270966	4042.930598
1	Random Forest	266.556824	944.265269

Predictive modeling reinforces the visual findings. The advanced random forest model substantially outperformed the baseline linear regression model, indicating that birth patterns are driven by multiple interacting factors rather than simple linear relationships. The model predicts that total births in 2025 will remain stable, supporting the conclusion that near-term maternity service demand is unlikely to decline sharply.

Overall, both exploratory analysis and predictive modeling suggest that Nebraska's maternity demand will remain steady in the near future. Hospitals should plan for stable or slightly fluctuating birth volumes rather than significant reductions in services.

Assumptions

This analysis assumes that aggregated birth data accurately reflects population-level trends and that reporting methods are consistent across all years. It also assumes that state-level averages are appropriate for healthcare planning decisions. Finally, the study assumes that short-term trends can be used to support near-future planning for maternity services.

Limitations

This analysis has several limitations. The data is aggregated and does not include individual-level records. Hospital-level data is not available, which limits local detail. The five-year time period restricts long-term forecasting, and external factors such as policy changes or economic conditions are not directly included.

Challenges

This project faced several challenges. It required careful alignment between national and state-level comparisons. Demographic changes had to be interpreted without making causal assumptions. Special care was taken to ensure ethical use of sensitive birth data and to avoid overgeneralizing results from short-term trends.

Future Uses / Additional Applications

This analysis can be expanded in the future if more data becomes available. County-level data could allow more detailed local analysis. Population projections could be added to support longer-term forecasting. The results may also be used for maternity workforce planning and for comparing Nebraska with neighboring states.

Recommendations

Hospitals should not reduce maternity services based only on national reports of declining births. State-level data should be reviewed each year to identify changes early.

Predictive models should be updated regularly as new data becomes available, and maternal demographic trends should be included in planning decisions.

Implementation Plan

Natality data should be updated each year. Predictive models should be re-run when new data is available. Results should be shared with hospital leaders and planners, and staffing and resource plans should be adjusted based on the updated projections.

Ethical Assessment

Birth data is highly sensitive and requires careful handling. This analysis uses only aggregated, publicly available data and avoids conclusions about individual behavior. The goal is system-level planning rather than personal judgment. Care is taken to avoid reinforcing stereotypes or misinterpreting demographic patterns.

Appendix

- Dataset
- Data cleaning code
- Visualization code

Audience Questions and Answers

1. Why focus on Nebraska instead of only national data?

Hospital planning happens at the state level, and national trends may not reflect local demand in Nebraska.

2. Why use data from 2020 to 2024?

Five years of data captures recent trends and allows short-term prediction while reflecting recent social and economic changes.

3. Why use birth counts instead of birth rates?

Birth counts are directly related to staffing, bed capacity, and maternity resources in hospitals.

4. Can five years of data support prediction?

Yes, it supports short-term prediction, but not long-term forecasting.

5. Why use both linear regression and random forest models?

Linear regression provides a simple baseline, while random forest improves accuracy by capturing complex patterns.

6. Why did the random forest model perform better?

Birth patterns are influenced by multiple interacting factors, which the random forest model can capture better.

7. What does the 2025 prediction mean for hospitals?

The prediction suggests stable birth volume, so hospitals should plan for steady maternity service demand.

8. Does this model predict individual births?

No, the model uses aggregated data and is intended for system-level planning only.

9. How were ethical concerns addressed?

Only public, aggregated data was used, and no individual behavior or personal outcomes were analyzed.

10. How can this analysis be improved in the future?

Adding more years of data, county-level detail, and population projections would improve accuracy.

Reference

Centers for Disease Control and Prevention. (2024). CDC WONDER natality database. U.S. Department of Health and Human Services. https://wonder.cdc.gov/natality-expanded-current.html